

2.6 Analysis of Functions I: Increase, Decrease, and Concavity

2.6.1 Increasing and decreasing functions

The terms increasing, decreasing, and constant are used to describe the behavior of a function as we travel left to right along its graph. For example, the function graphed in Figure 2.1 can be described as increasing to the left of $x = 0$, decreasing from $x = 0$ to $x = 2$, increasing from $x = 2$ to $x = 4$, and constant to the right of $x = 4$.

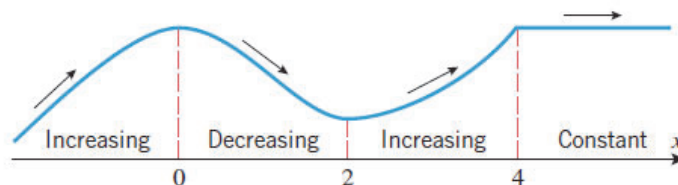


Figure 2.1

The following definition, which is illustrated in Figure 2.2, expresses these intuitive ideas precisely.

Definition 2.6.1

Let f be defined on an interval, and let x_1 and x_2 denote points in that interval.

- (a) f is **increasing** on the interval if $f(x_1) < f(x_2)$ whenever $x_1 < x_2$.
- (b) f is **decreasing** on the interval if $f(x_1) > f(x_2)$ whenever $x_1 < x_2$.
- (c) f is **constant** on the interval if $f(x_1) = f(x_2)$ for all points x_1 and x_2 .

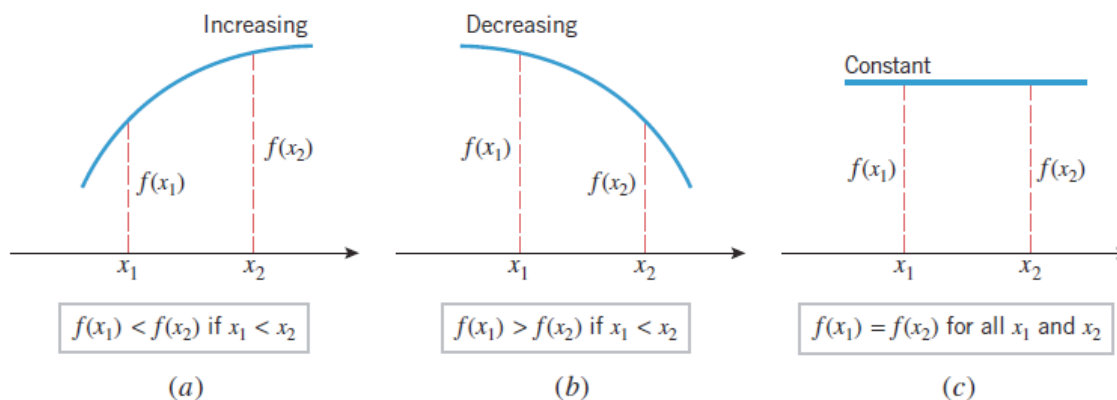


Figure 2.2

Figure 2.3 suggests that a differentiable function f is increasing on any interval where each tangent line to its graph has positive slope, is decreasing on any interval where each tangent line to its graph has negative slope, and is constant on any interval where each tangent line to its graph has zero slope.

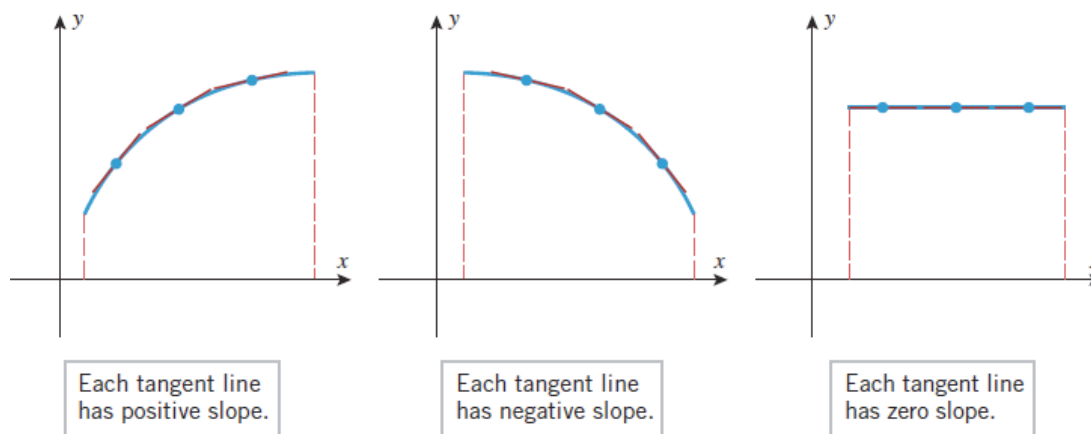


Figure 2.3

Theorem 2.6.1

Let f be a function that is continuous on a closed interval $[a, b]$ and differentiable on the open interval (a, b) .

- (a) If $f'(x) > 0$ for every value of x in (a, b) , then f is increasing on $[a, b]$.
- (b) If $f'(x) < 0$ for every value of x in (a, b) , then f is decreasing on $[a, b]$.
- (c) If $f'(x) = 0$ for every value of x in (a, b) , then f is constant on $[a, b]$.

Although stated for closed intervals, Theorem 2.6.1 is applicable on any interval on which f is continuous.

Example 2.6.1

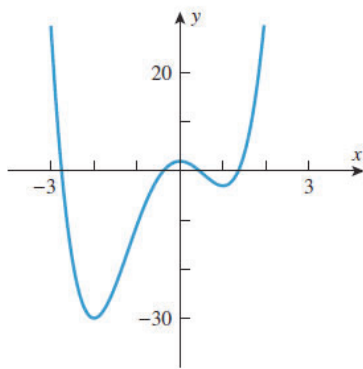
Find the intervals on which $f(x) = x^2 - 3x + 8$ is increasing and the intervals on which it is decreasing.

Example 2.6.2

Find the intervals on which $f(x) = x^5$ is increasing and the intervals on which it is decreasing.

Example 2.6.3

- (a) Use the graph of $f(x) = 3x^4 + 4x^3 - 12x^2 + 2$ in Figure 2.4 to make a conjecture about the intervals on which f is increasing or decreasing.
- (b) Use Theorem 2.11.1 to determine whether your conjecture is correct.



$$f(x) = 3x^4 + 4x^3 - 12x^2 + 2$$

Figure 2.4

2.6.2 Concavity

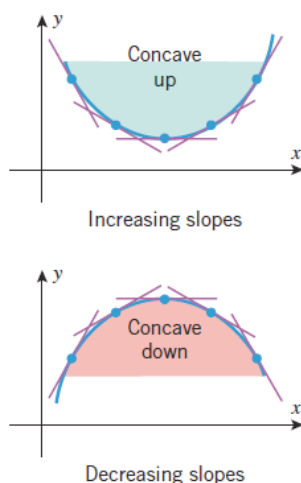


Figure 2.5

Figure 2.5 suggests two ways to characterize the concavity of a differentiable function f on an open interval:

- f is concave up on an open interval if its tangent lines have increasing slopes on that interval and is concave down if they have decreasing slopes.
- f is concave up on an open interval if its graph lies above its tangent lines on that interval and is concave down if it lies below its tangent lines.

Definition 2.6.2

If f is differentiable on an open interval, then f is said to be **concave up** on the open interval if f' is increasing on that interval, and f is said to be **concave down** on the open interval if f' is decreasing on that interval.

Since the slopes of the tangent lines to the graph of a differentiable function f are the values of its derivative f' , it follows from Theorem 2.6.1 (applied to f' rather than f). Thus, we have the following theorem.

Theorem 2.6.2

Let f be twice differentiable on an open interval.

- If $f''(x) > 0$ for every value of x in the open interval, then f is concave up on that interval.
- If $f''(x) < 0$ for every value of x in the open interval, then f is concave down on that interval.

Example 2.6.4

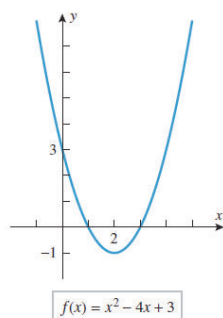


Figure 2.6

Figure 2.6 suggests that the function $f(x) = x^2 - 4x + 3$ is concave up on the interval $(-\infty, +\infty)$. This is consistent with Theorem 2.6.2, since $f'(x) = 2x - 4$ and $f''(x) = 2$, so

$$f''(x) > 0 \quad \text{on the interval } (-\infty, +\infty)$$

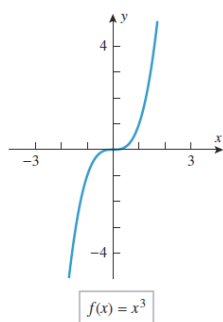


Figure 2.7

Also, Figure 2.7 suggests that $f(x) = x^3$ is concave down on the interval $(-\infty, 0)$ and concave up on the interval $(0, +\infty)$. This agrees with Theorem 2.6.2, since $f'(x) = 3x^2$ and $f''(x) = 6x$, so

$$f''(x) < 0 \quad \text{if } x < 0 \quad \text{and} \quad f''(x) > 0 \quad \text{if } x > 0$$

2.6.3 Inflection points

We see from Figure 2.7 that the graph of $f(x) = x^3$ changes from concave down to concave up at $x = 0$. Points where a curve changes from concave up to concave down or vice versa are of special interest, so there is some terminology associated with them.

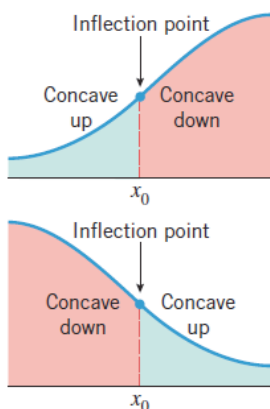


Figure 2.8

Definition 2.6.3

If f is continuous on an open interval containing a value x_0 , and if f changes the direction of its concavity at the point $(x_0, f(x_0))$, then we say that f has an **inflection point at x_0** , and we call the point $(x_0, f(x_0))$ on the graph of f an **inflection point of f** (Figure 2.8).

Example 2.6.5

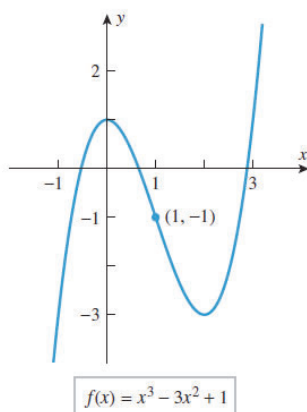


Figure 2.9

Figure 2.9 shows the graph of the function $f(x) = x^3 - 3x^2 + 1$. Use the first and second derivatives of f to determine the intervals on which f is increasing, decreasing, concave up, and concave down. Locate all inflection points and confirm that your conclusions are consistent with the graph.

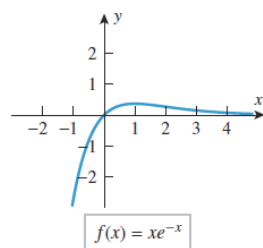
Example 2.6.6

Figure 2.10

Figure 2.10 suggests that the function $f(x) = xe^{-x}$ has an inflection point but its exact location is not evident from the graph in this figure. Use the first and second derivatives of f to determine the intervals on which f is increasing, decreasing, concave up, and concave down. Locate all inflection points.

Solution. Calculating the first two derivatives of f we obtain (verify)

$$f'(x) = (1 - x)e^{-x}$$

$$f''(x) = (x - 2)e^{-x}$$

Keeping in mind that e^{-x} is positive for all x , the sign analysis of these derivatives is easily determined:

Interval	$(1 - x)(e^{-x})$	$f'(x)$	Conclusion
$x < 1$	$(+)(+)$	$+$	f is increasing on $(-\infty, 1]$
$x > 1$	$(-)(+)$	$-$	f is decreasing on $[1, +\infty)$

Interval	$(x - 2)(e^{-x})$	$f''(x)$	Conclusion
$x < 2$	$(-)(+)$	$-$	f is concave down on $(-\infty, 2)$
$x > 2$	$(+)(+)$	$+$	f is concave up on $(2, +\infty)$

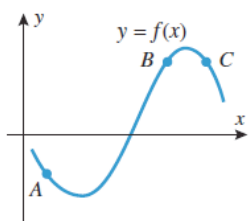
The second table shows that there is an inflection point at $x = 2$, since f changes from concave down to concave up at that point. All of these conclusions are consistent with the graph of f .

Example 2.6.7

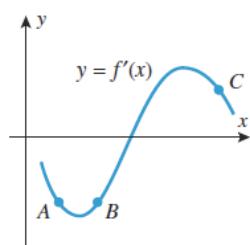
Find the inflection points, if any, of $f(x) = x^6$

2.6.4 Exercises

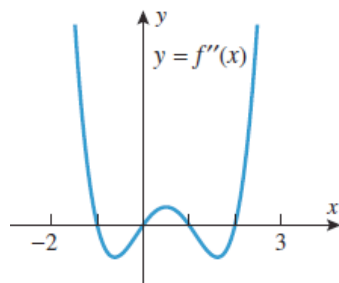
1. In each part, sketch the graph of a function f with the stated properties, and discuss the signs of f' and f'' .
 - 1.1. The function f is concave up and increasing on the interval $(-\infty, +\infty)$.
 - 1.2. The function f is concave down and increasing on the interval $(-\infty, +\infty)$.
 - 1.3. The function f is concave up and decreasing on the interval $(-\infty, +\infty)$.
 - 1.4. The function f is concave down and decreasing on the interval $(-\infty, +\infty)$.
2. In each part, sketch the graph of a function f with the stated properties.
 - 2.1. f is increasing on $(-\infty, +\infty)$, has an inflection point at the origin, and is concave up on $(0, +\infty)$.
 - 2.2. f is increasing on $(-\infty, +\infty)$, has an inflection point at the origin, and is concave down on $(0, +\infty)$.
 - 2.3. f is decreasing on $(-\infty, +\infty)$, has an inflection point at the origin, and is concave up on $(0, +\infty)$.
 - 2.4. f is decreasing on $(-\infty, +\infty)$, has an inflection point at the origin, and is concave down on $(0, +\infty)$.



3. Use the graph of the equation $y = f(x)$ in the accompanying figure to find the signs of dy/dx and d^2y/dx^2 at the points A, B, and C.

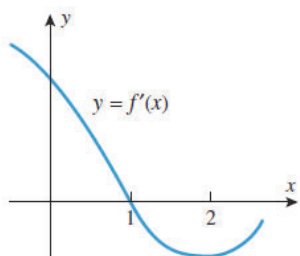


4. Use the graph of the equation $y = f'(x)$ in the accompanying figure to find the signs of dy/dx and d^2y/dx^2 at the points A, B, and C.



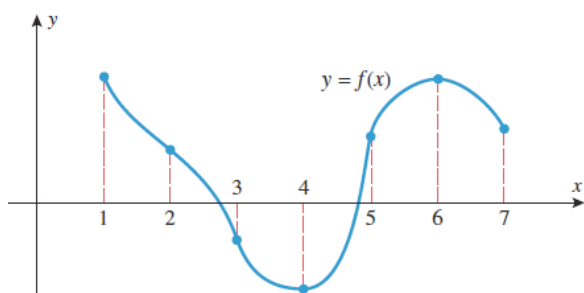
5. Use the graph of $y = f''(x)$ in the accompanying figure to determine the x -coordinates of all inflection points of f . Explain your reasoning.

6. Use the graph of $y = f'(x)$ in the accompanying figure to replace the question mark with $<$, $=$, or $>$, as appropriate. Explain your reasoning.



- 6.1. $f(0) ? f(1)$
 6.2. $f(1) ? f(2)$
 6.3. $f'(0) ? 0$
 6.4. $f'(1) ? 0$
 6.5. $f''(0) ? 0$
 6.6. $f''(2) ? 0$

7. In each part, use the graph of $y = f(x)$ in the accompanying figure to find the requested information.



- 7.1. Find the intervals on which f is increasing.
 7.2. Find the intervals on which f is decreasing.
 7.3. Find the open intervals on which f is concave up.
 7.4. Find the open intervals on which f is concave down.
 7.5. Find all values of x at which f has an inflection point.

8. Use the graph in Exercise 7 to make a table that shows the signs of f' and f'' over the interval $(1, 2)$, $(2, 3)$, $(3, 4)$, $(4, 5)$, $(5, 6)$, and $(6, 7)$.
9. A sign chart is presented for the first and second derivatives of a function f . Assuming that f is continuous everywhere, find: (a) the intervals on which f is increasing, (b) the intervals on which f is decreasing, (c) the open intervals on which f is concave up, (d) the open intervals on which f is concave down, and (e) the x -coordinates of all inflection points.

9.1. _____

Interval	sign of $f'(x)$	sign of $f''(x)$
$x < 1$	—	+
$1 < x < 2$	+	+
$2 < x < 3$	+	—
$3 < x < 4$	—	—
$4 < x$	—	+

9.2. _____

Interval	sign of $f'(x)$	sign of $f''(x)$
$x < 1$	+	+
$1 < x < 3$	+	—
$3 < x$	+	+

10. Find: (a) the intervals on which f is increasing, (b) the intervals on which f is decreasing, (c) the open intervals on which f is concave up, (d) the open intervals on which f is concave down, and (e) the x -coordinates of all inflection points.

10.1. $f(x) = 5 - 4x - x^2$

10.8. $f(x) = \sqrt[3]{x^2 + x + 1}$

10.2. $f(x) = (2x + 1)^3$

10.9. $f(x) = x^{4/3} - x^{1/3}$

10.3. $f(x) = 5 + 12x - x^3$

10.10. $f(x) = (x^{2/3} - 1)^2$

10.4. $f(x) = 3x^4 - 4x^3$

10.11. $f(x) = x^{2/3} - x$

10.5. $f(x) = x^4 - 5x^3 + 9x^2$

10.12. $f(x) = e^{-x^2/2}$

10.6. $f(x) = \frac{x - 2}{(x^2 - x + 1)^2}$

10.13. $f(x) = xe^{x^2}$

10.14. $f(x) = \ln \sqrt{x^2 + 4}$

10.7. $f(x) = \frac{x}{x^2 + 2}$

10.15. $f(x) = x^3 \ln x$

11. In each part sketch a continuous curve $y = f(x)$ with the stated properties.

11.1. $f(2) = 4, f'(2) = 0, f''(x) > 0$ for all x

11.2. $f(2) = 4, f'(2) = 0, f''(x) < 0$ for all $x < 2, f''(x) > 0$ for all $x > 2$

11.3. $f(2) = 4, f''(x) < 0$ for $x \neq 2$ and $\lim_{x \rightarrow 2^+} f'(x) = +\infty, \lim_{x \rightarrow 2^-} f'(x) = -\infty$